

Representing Integers on the Number Line

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A **number line** is a straight line used to represent numbers in order.

The Origin

The **origin** is 0, which separates **positive integers** and **negative integers**.

$$\dots - 2 \quad - 1 \quad \boxed{0} \quad 1 \quad 2 \dots$$

Positive Integers

Positive integers are located to the **right** of 0.

0 1 2 3 4 $\longrightarrow$

Negative Integers

Negative integers are located to the **left** of 0.

← -4 - 3 - 2 - 1 0

Greater or Smaller?

The farther a number is to the **right**, the **greater** its value.

The farther a number is to the **left**, the **smaller** its value.

−5	−4	−3	−2	−		
1	0	1	2	3	4	5

smaller ← ————— → *greater*

Unique Positions

Each integer has a **unique position** on the number line.

No two integers can occupy the same spot.

Why It Matters

Locating integers on the number line helps:

- ▶ Compare numbers
- ▶ Understand their order

Examples

Example 1

Plot -3 , 0 , and 4 on a number line from -5 to 5 .



Example 2

Which integer is located 3 units to the left of 0?

0, -1 , -2 , -3

Counting 3 steps left from 0 lands on $\boxed{-3}$.

Example 3

What integer is halfway between -8 and 0 ?

$$\frac{-8 + 0}{2} = -4$$

Halfway between -8 and 0 is $\boxed{-4}$.

Example 4

List the integers from left to right:

2, -4 , 0, -1 , 3.

-4 , -1 , 0, 2, 3

Smallest numbers go on the left, largest on the right.

Practice Exercises

Practice 1

Draw a number line from -6 to 6 . Plot the integers -4 , 0 , and 5 .

Practice 2

Draw a number line from -8 to 8 . Plot the integers -7 , -2 , and 4 .

Practice 3

Draw a number line from -10 to 10 . Plot the integers -9 , 1 , and 8 .

Practice 4

Draw a number line from -5 to 5 . Plot the integers -3 , 2 , and 5 .

Practice 5

Draw a number line from -12 to 12 . Plot the integers -10 , 0 , and 9 .

Practice 6

Which integer is located 4 units to the left of 0?

Practice 7

Which integer is located 6 units to the right of 0?

Practice 8

What integer is halfway between -8 and 0 on the number line?

Practice 9

List the integers from left to right:

3, -5 , 0, -1 , 4.

Practice 10

List the integers from left to right:

$-7, 2, -3, 5, 0.$

Activities

Activity 1

Draw a number line from -7 to 7 . Plot the integers -5 , 0 , and 6 .

Activity 2

Draw a number line from -9 to 9 . Plot the integers -8 , -1 , and 7 .

Activity 3

Draw a number line from -11 to 11 . Plot the integers -10 , 2 , and 9 .

Activity 4

Draw a number line from -6 to 6 . Plot the integers -4 , 3 , and 6 .

Activity 5

Draw a number line from -13 to 13 . Plot the integers -11 , 0 , and 10 .

Activity 6

Which integer is located 5 units to the left of 0?

Activity 7

Which integer is located 7 units to the right of 0?

Activity 8

What integer is halfway between -10 and 0 on the number line?

Activity 9

List the integers from left to right:

2, -6 , 1, -2 , 5.

Activity 10

List the integers from left to right:

$-8, 3, -4, 6, 0.$

Thank you!